

Project Design Phase-II

Technology Stack (Architecture & Stack)

Date: 30 September 2026

Team ID: SWTID-2026-9911

Project Name: Auto Ticket Classification using Flow Designer

Team Size: 4

Team Leader: Indhu T

Team Members: Mahalakshmi M; Venmathi S; Amala Prathiba J

Maximum Marks: 4 Marks

=====

## TECHNICAL ARCHITECTURE

=====

Project Objective:

Automate the ticket classification workflow in ServiceNow for the school IT helpdesk. The application allows students and teachers to submit incident requests (e.g., Wi-Fi issues, projector failures, password problems, slow computers), analyzes keywords in the short description and description fields using Flow Designer, automatically assigns the correct category and IT assignment group, and sends status notifications with minimal manual effort.

Architecture Diagram Flow:

[User: Student / Teacher]

|



[ServiceNow Service Portal / Incident Request Form]

|



[Automation Layer: Flow Designer / Business Rules]

|



[Classification & Assignment Logic (Keyword Parsing)]

|



[Data Layer: ServiceNow Incident Tables]

|



[Output: Category Assignment / Notifications / Reports]

Deployed Platform:

The solution is deployed on the ServiceNow cloud platform.

ServiceNow Data Tables & Components:

- USER: Student / Teacher / Requester
- SERVICENOW SERVICE PORTAL: Incident Request Form (Short Description & Description)
- AUTOMATION LAYER: Flow Designer, Business Rules, Script Includes
- APPROVAL & ROUTING: IT Support Assignment Groups
- DATA LAYER: Incident [incident] Table, Category Table, Work Notes/Logs
- OUTPUT: Automated Category, Email Notifications, IT Helpdesk Dashboard

=====

TABLE 1: TECHNOLOGY STACK / ARCHITECTURE COMPONENTS

| =====             |                               |                                                                                                                                                |
|-------------------|-------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------|
| S.No.             | Component                     | Description                                                                                                                                    |
| Technology        |                               |                                                                                                                                                |
| -----+-----+----- |                               |                                                                                                                                                |
| +-----            |                               |                                                                                                                                                |
| 1                 | User Interface                | Student and teacher-facing incident reporting form and Helpdesk Portal.<br>  ServiceNow Service Portal / Incident Form / Workspace             |
| 2                 | Application Logic-1           | Workflow logic for keyword parsing, automatic category assignment,<br>and task routing.   Flow Designer                                        |
| 3                 | Application Logic-2           | Server-side logic for keyword pattern matching and automated field<br>validation where required.   Business Rules, Script Includes, Glide APIs |
| 4                 | Application Logic-3           | Priority calculation and assignment group routing logic.<br>  Assignment Rules, Flow Designer                                                  |
| 5                 | Database                      | Stores incident tickets, categories, user details, assignment rules, and logs.<br>  ServiceNow platform database / Application tables          |
| 6                 | Cloud Database                | Managed cloud data layer provided by the ServiceNow platform.<br>  ServiceNow Cloud Platform                                                   |
| 7                 | File Storage                  | Optional storage for screenshot attachments or supporting error logs<br>submitted by users.   ServiceNow Attachments / Platform file storage   |
| 8                 | External API-1                | No mandatory third-party API required; integration can be added for<br>external IT alerting.   ServiceNow REST API / IntegrationHub (Optional) |
| 9                 | External API-2                | Optional integration with school communication tools (e.g., Teams/Email<br>notifications).   REST/SOAP APIs, IntegrationHub                    |
| 10                | Machine Learning Model        | Optional Predictive Intelligence (Keyword/Text Analytics using<br>Flow Designer logic).   ServiceNow Predictive Intelligence / Flow Designer   |
| 11                | Infrastructure (Server/Cloud) | Application is hosted on the ServiceNow cloud platform; no local<br>infrastructure required.   ServiceNow Cloud / SaaS Platform                |

=====

TABLE 2: APPLICATION CHARACTERISTICS

=====

| S.No. | Characteristics          | Description                                                                                                                                                                    |
|-------|--------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|       | Technology               |                                                                                                                                                                                |
| <hr/> |                          |                                                                                                                                                                                |
| 1     | Open-Source Frameworks   | Uses the native ServiceNow platform. JavaScript is used for server-side scripting and keyword matching logic.   JavaScript / ServiceNow APIs                                   |
| 2     | Security Implementations | Role-based access control (RBAC), ACLs, authentication, and secure access to student/teacher data and IT incident records.   ServiceNow ACLs, Roles, Authentication, HTTPS/TLS |
| 3     | Scalable Architecture    | Cloud-based SaaS architecture handles high volumes of daily incident tickets without maintaining local servers.   ServiceNow Cloud Platform, Flow Designer                     |
| 4     | Availability             | Runs continuously on ServiceNow cloud services, enabling 24/7 access for students and teachers across the campus.   ServiceNow Cloud / SaaS                                    |
| 5     | Performance              | Instant classification occurs upon ticket creation using efficient Flow Designer triggers without delays.   Flow Designer, Business Rules, Glide APIs                          |

=====

CORE AUTOMATED FLOW

=====

- Incident Submission : Student/Teacher fills out the Incident form with Short Description and Description.
- Trigger : Flow Designer triggers immediately upon new ticket creation.

3. Keyword Analysis : System scans Short Description & Description for target keywords (e.g., "Wi-Fi", "projector", "password", "slow").
4. Auto-Classification : Category, Subcategory, and Priority are assigned automatically.
5. Task Assignment : Incident is directly routed to the corresponding IT Support Group.
6. Notification : Requester receives automated notification with ticket number and assigned category.
7. Resolution & Closure: IT staff resolves the issue, updates resolution details, and closes the ticket.

=====

#### PROJECT TEAM DETAILS

=====

| Role | Name | Team ID |
|------|------|---------|
|------|------|---------|

|        |        |       |
|--------|--------|-------|
| -----+ | -----+ | ----- |
|--------|--------|-------|

|             |         |                 |
|-------------|---------|-----------------|
| Team Leader | Indhu T | SWTID-2026-9911 |
|-------------|---------|-----------------|

|             |               |                 |
|-------------|---------------|-----------------|
| Team Member | Mahalakshmi M | SWTID-2026-9911 |
|-------------|---------------|-----------------|

|             |            |                 |
|-------------|------------|-----------------|
| Team Member | Venmathi S | SWTID-2026-9911 |
|-------------|------------|-----------------|

|             |                  |                 |
|-------------|------------------|-----------------|
| Team Member | Amala Prathiba J | SWTID-2026-9911 |
|-------------|------------------|-----------------|

=====

#### REFERENCES

=====

- ServiceNow Documentation: Flow Designer, Incident Management, and Data Lookup Rules.[cite: 2]

- ServiceNow Developer Documentation: JavaScript APIs, GlideRecord, and Platform Architecture.[cite: 2]